

8 Realistic blinking

The eyes need to close only briefly and stay open for a while between blinks. Add this function, `blink()`, under the code you typed in Step 7. It blinks the eyes for a quarter of a second (250 milliseconds), then finishes with a command that tells `mainloop()` to call it again after 3 seconds (3,000 milliseconds).

```
c.itemconfigure(eye_right, fill=new_color)

def blink():
    toggle_eyes()
    root.after(250, toggle_eyes)
    root.after(3000, blink)

root = Tk()
```

Close the eyes.

Wait 250 milliseconds, then open the eyes.

Wait 3,000 milliseconds, then blink again.

9 Animate!

Put this line in the main part of your program, just above the last line. Now run the program. Your pet will come to life after 1 second (1,000 milliseconds) and sit there blinking until you close the window.

```
root.after(1000, blink)
root.mainloop()
```

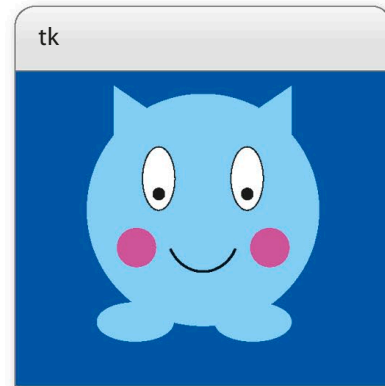
Wait 1,000 milliseconds, then start blinking.

Changing moods

Screen Pet looks quite happy just now, with its little smile, but let's cheer it up even more. We'll give it a bigger, beaming smile and bright, rosy cheeks.

10 Make a happy face

Add this code to the part of the program that draws Screen Pet, after the line that creates the "normal" mouth. As well as a happy mouth and pink cheeks, it also draws a sad mouth. They will all remain hidden for now.



Create a happy mouth.

Create a sad mouth.

```
mouth_normal = c.create_line(170, 250, 200, 272, 230, 250, smooth=1, width=2, state=NORMAL)
mouth_happy = c.create_line(170, 250, 200, 282, 230, 250, smooth=1, width=2, state=HIDDEN)
mouth_sad = c.create_line(170, 250, 200, 232, 230, 250, smooth=1, width=2, state=HIDDEN)
```

```
cheek_left = c.create_oval(70, 180, 120, 230, outline='pink', fill='pink', state=HIDDEN)
cheek_right = c.create_oval(280, 180, 330, 230, outline='pink', fill='pink', state=HIDDEN)
```

```
c.pack()
```

These lines create pink, blushing cheeks.